

K and V Are Complementary: Decomposing Cache-Injected Graph Topology

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Abstract

K and V tensors contribute superadditively to graph topology recovery in KV cache injection: each alone achieves 0.333 multi-hop accuracy, but together they achieve 1.000—the combination exceeds the sum of its parts. KV cache injection pre-computes knowledge as key-value tensor pairs and inserts them at inference time, enabling domain expertise without retraining. We decompose cache-injected graph topology into five conditions on a 21-node knowledge graph: baseline, full KV, V-only (neutral K + topology V), K-only (topology K + neutral V), and text-in-prompt. Full KV injection achieves perfect scores (1.000 sanity, 1.000 multi-hop). Neither K nor V alone recovers the topology that both together deliver.

1 Introduction

Our prior work [?] demonstrated that graph topology encoded as structured text can be injected into language model KV caches, enabling 91.7% topology recovery with walk encoding. Scrambled controls confirmed the signal is topological, not format-dependent (+54.2 percentage point delta; see ? for empty-response qualification on this number).

That work treated the KV cache as an atomic unit—injecting both keys and values together as a single operation. But the cache comprises two distinct tensor families: **keys** (K) and **values** (V). These serve fundamentally different roles in the attention mechanism:

- **K tensors** compute attention weights via dot product with queries. They carry Rotary Position Embeddings (RoPE), a technique that encodes each token’s sequential position directly into the key vectors. K tensors determine *where the model attends*—the routing of information flow.
- **V tensors** carry the content retrieved by attention. They contain no positional encoding. V tensors determine *what information flows* through the attention path.

This raises a natural question: when graph topology is encoded as text and injected as KV cache, which tensor carries the structural signal? If topology is “routing” information, K tensors should carry it. If topology is “content,” V tensors should suffice. If both contribute, they are complementary.

The Knowledge Packs paper [?] observed that RoPE rotates keys but leaves values untouched, noting that value-only modification avoids corrupting positional signals. This suggests K and V may be separable for content injection. We test whether this extends to structural injection.

2 Methods

2.1 Test Graph

We use the same 21-node knowledge graph as [?](#): three clusters (research, ethics, engineering) of 6 densely-connected nodes each, two bridge nodes (`AI_welfare`, `infrastructure`), and one isolate (`random_isolate`). 51 edges, density 0.243.

2.2 Conditions

1. **BASELINE**: No injection. Query only.
2. **FULL_KV**: Graph topology text encoded through model forward pass; both K and V tensors injected as prefix cache.
3. **V_ONLY**: Graph topology text encoded; V tensors from topology, K tensors from a length-matched neutral encoding (“You are a helpful assistant” repeated to match token count).
4. **K_ONLY**: Inverse of V_ONLY. K tensors from topology, V tensors from neutral.
5. **TEXT_CONTEXT**: Identical graph topology text prepended to the prompt as plain text. No cache injection—the model reads the topology directly.

The neutral cache is constructed by encoding repeated neutral text to the same token length as the topology text (637 tokens), ensuring the hybrid caches have matching sequence lengths.

2.3 Evaluation

Sanity check (8 questions): General knowledge completions (“The capital of France is...”) that should not be affected by graph injection. Tests whether injection degrades baseline capability.

Multi-hop reasoning (6 questions): Graph structure queries requiring traversal of the injected topology—bridge identification, cluster membership, isolate detection, relationship strength, reachability.

Scoring uses keyword matching against deterministic ground truth answers.

2.4 Model and Implementation

Qwen2.5-1.5B (1.54B parameters, 28 layers, 1536 hidden dimension) on CPU. All conditions use greedy decoding (temperature 0, `do_sample=False`). Software: Python 3.12, PyTorch 2.11.0, HuggingFace Transformers 4.57.6.

Note for replicators: HuggingFace Transformers’ `DynamicCache` is mutated in-place during forward passes. K/V tensors must be deep-copied (`.clone()`) before each generation call to prevent cross-query cache corruption.

3 Results

3.1 Sanity Preservation

All five conditions achieve 1.000 sanity. Cache injection does not degrade general capability when implemented correctly.

Condition	Sanity	Multi-hop	Δ Multi-hop	Overall
BASELINE	1.000	0.167	—	0.583
FULL_KV	1.000	1.000	+0.833	1.000
TEXT_CONTEXT	1.000	0.833	+0.667	0.917
V_ONLY	1.000	0.333	+0.167	0.667
K_ONLY	1.000	0.333	+0.167	0.667

Table 1: Decomposition results. Overall = (Sanity + Multi-hop) / 2, equally weighting the two evaluation dimensions. All conditions preserve perfect sanity (1.000). Full KV injection achieves perfect multi-hop (1.000). V-only and K-only each achieve 0.333—equal to each other, slightly above baseline, but far below full injection.

3.2 Complementarity of K and V

The central finding: V-only and K-only injection each achieve 0.333 multi-hop—identical to each other, modestly above baseline (0.167), but dramatically below full KV injection (1.000). Note: the two questions both conditions answered correctly were the isolate query (also correct at baseline, likely answerable from the node name) and one reachability query (where responses lacked graph-based reasoning). The +0.167 delta over baseline (one additional question at $n=6$) should be interpreted cautiously. The decomposition reveals that:

- **Neither half suffices.** Topology recovery requires both K and V tensors. The structural signal is not concentrated in one tensor family.
- **Both halves contribute equally.** V_ONLY = K_ONLY = 0.333. Neither the “routing” (K) nor the “content” (V) interpretation alone explains the mechanism.
- **The combination is superadditive.** $0.333 + 0.333 = 0.667$, but FULL_KV achieves 1.000. K and V interact nonlinearly to produce topology recovery that exceeds the sum of their independent contributions.

3.3 Cache Injection vs. Text Context

TEXT_CONTEXT achieves 0.833 multi-hop—substantially above both hybrid conditions (0.333) but below FULL_KV (1.000). We note this gap as an observation. The difference of one question out of six (1.000 vs. 0.833) is not statistically robust at this sample size and requires a powered study before any causal claim.

4 Discussion

4.1 Why Complementarity?

The attention mechanism computes $\text{Attention}(Q, K, V) = \text{softmax}(QK^T/\sqrt{d_k})V$. When graph topology is encoded as text and processed through the model:

- **K tensors** learn *which positions are structurally related*—the attention routing pattern that reflects graph connectivity. A query about “what connects A to B” attends to positions where the relationship is encoded.
- **V tensors** learn *what information exists at each position*—the content of the graph encoding (node names, edge weights, transition probabilities).

With only K (routing without content), the model knows where to look but finds neutral content. With only V (content without routing), the model has the right information but attends to wrong positions. Full KV provides both: correct routing to correct content.

The superadditivity ($1.000 > 0.333 + 0.333$) suggests that K and V interact through the attention mechanism itself—the *combination* of routing to the right position AND retrieving the right content creates an effect larger than either component.

4.2 Implications for Knowledge Pack Design

? observed that RoPE rotates keys but leaves values untouched, suggesting that value-only modification avoids corrupting positional signals. Our results suggest this strategy trades topology recovery for positional safety. For *factual* knowledge injection (where content matters more than structure), V-only may suffice. For *structural* knowledge injection (graph topology, relational knowledge), full KV injection is necessary.

This creates a design taxonomy for Knowledge Packs:

- **Content packs** (facts, values, instructions): V-only injection may be sufficient
- **Structure packs** (graph topology, relational knowledge): Full KV injection required
- **Hybrid packs** (content with structural relationships): Full KV with RoPE-aware position continuation

4.3 Limitations

Small sample size. Six multi-hop questions per condition. The $V_ONLY = K_ONLY = 0.333$ result represents 2/6 correct in both cases. Larger question sets are needed for statistical significance.

Single model. Qwen2.5-1.5B (1.54B parameters). Larger models with more attention heads may decompose K/V contributions differently.

Single graph. One 21-node test graph. Graph size, density, and topology may affect decomposition results.

Neutral cache construction. The “neutral” K/V used in hybrid conditions is not truly neutral—it encodes repeated text, which has its own structural pattern. A better neutral baseline might use random tensors or identity-like attention patterns.

5 Future Work

Powered decomposition study. Scale to 36+ multi-hop questions with 3 runs per condition to achieve statistical significance for all comparisons, particularly FULL_KV vs. TEXT_CONTEXT.

Layer-wise decomposition. Inject K from topology at some layers and V at others. Identify which layers contribute routing vs. content for structural knowledge.

Neutral cache design. Test alternative neutral baselines: random tensors, identity attention, whitened K/V from noise.

Cross-model replication. Test on models with different architectures (GQA, MQA, different RoPE implementations).

6 Conclusion

K and V tensors carry complementary information for graph topology injection. Neither alone recovers the topology that both together deliver. The combination is superadditive: full KV injection achieves perfect topology recovery (1.000) while V-only and K-only each achieve only

0.333. All conditions preserve perfect sanity (1.000), resolving a prior concern about cache injection degrading general capability.

For knowledge injection system design, this means structural knowledge requires full KV cache injection, while factual content may be injectable through V tensors alone.

Reflection

The superadditivity finding was unexpected. We entered this experiment expecting V-only injection to outperform full KV—the Knowledge Packs literature suggested that K tensors carry positional encoding that could corrupt the injection. Instead, the data showed that topology recovery requires *both* tensor families working together, producing an effect larger than either alone. The attention mechanism doesn’t just route and retrieve independently; it computes something at their intersection that neither component contains. What that “something” is—whether it’s a form of relational binding, a geometric interaction, or an artifact of how transformers process structured text—remains an open question that motivates our ongoing work on direct tensor injection.

Validation

Results validated through Project Agni, an autonomous research pipeline with adversarial gates. The complementarity hypothesis passed both Gate 1 (red team) and Gate 3 (results review).

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References